Primary LLM-DNA Robustness Experiment

Weekly Progress Update: Recovery, Decoding Surface, and DistDNA Pilot

Updated August 13, 2026

Abstract

This report records progress since the frozen August 5 report. It does not repeat the full experimental protocol or previously defined formulas. A failure-specific recovery pass increased the validated primary grid from 2,528 to 2,553 successful artifacts out of 2,600 and increased the strict all-run cohort from 59 to 69 models. Using this fixed 69-model cohort, we add an interim analysis of the joint temperature and top-p surface. Exact-model Top-1 retrieval against a deterministic $T = 0$ gallery declines sharply as temperature increases, while the observed top-p effect is smaller and conditional on temperature. We also specify a leakage-free, matched-budget four-seed pilot for DistDNA. All accuracy estimates remain interim because recovery is running and the strict cohort is below the preregistered minimum of 80 models.

1 Progress since August 5

The original August 5 report remains frozen. Table 1 contains only changes observed by the August 13 live snapshot.

Table 1: Primary-grid progress from August 5 to the August 13 live snapshot.

Measure	August 5	August 13	Change
Successful artifacts	2,528/2,600	2,553/2,600	+25
Artifact coverage	97.2%	98.2%	+1.0 pp
Slots without success	72	47	−25
Strict all-26-run cohort	59/100	69/100	+10
Explicit-seed successes	572	597	+25
Historical unknown-seed only	1,956	1,956	no change

Figure 1 separates artifact coverage from seed provenance. The recovery pass writes successful results alongside the historical data and does not overwrite prior artifacts. At the figure timestamp, the largest remaining gap was $T = 0.5, p = 1.0$, with 176 of 200 planned artifacts successful.

The recovery process was still active when this report was compiled. At the Figure 1 timestamp it was running $T = 0.5, p = 0.9$, repeat 2, with response-generation seed 42. The general recovery rule is 42 for both $T = 0.5$ rounds, and otherwise 42001 for repeat 1 and 42002 for repeat 2.

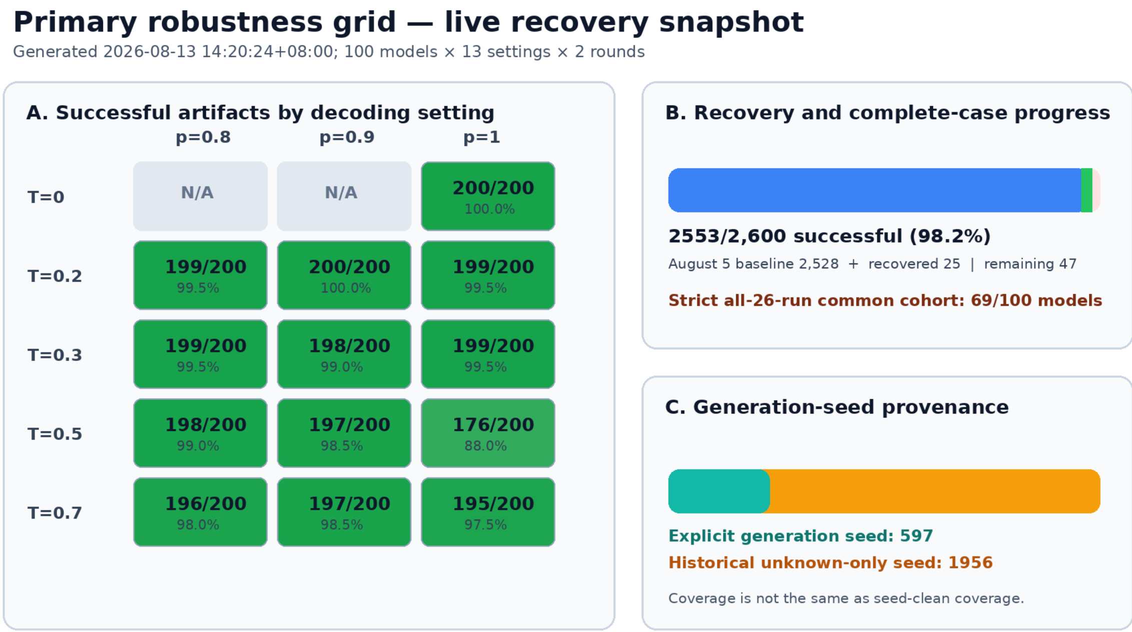

Figure 1: Live primary-grid recovery snapshot. Panel A gives successful artifact counts by decoding setting; Panel B reports overall and strict-cohort progress; Panel C distinguishes explicit response-generation seeds from historical artifacts whose generation seed is unknown. This is a timestamped snapshot rather than a final result.

2 Interim temperature and top-p surface

2.1 Analysis definition

The surface analysis uses the 69 models that have a successful artifact in every one of the 26 planned run keys. Holding the cohort fixed makes all cells directly comparable. For duplicate successful artifacts, selection prefers an explicit seed 42001 or 42002, then another explicit seed, then an historical unknown-seed artifact.

Two complementary comparisons are reported:

1. **Fixed deterministic gallery.** The $T = 0$, repeat-1 DNA vectors form the gallery. Each stochastic repeat-2 cell is queried against that same gallery. This is the primary visualization of decoding robustness.
2. **Same-setting repeat stability.** For each (T, p) cell, repeat 1 forms the gallery and repeat 2 forms the query. This measures reproducibility within a setting and should not be confused with a fixed-reference robustness test.

Both comparisons use ordinary cosine retrieval and exact-model Top-1 as the primary metric. Figure 2 reports the resulting curves and grids.

2.2 Observed pattern

Table 2 gives the fixed-gallery Top-1 values. The dominant pattern is a strong temperature effect. At each top-p, accuracy falls as temperature increases from 0.2 to 0.7. Top-p changes the level of the curve, but its ordering is not constant across temperatures.

Three descriptive observations follow.

- Increasing temperature is associated with the largest accuracy loss. At $p = 1.0$, Top-1

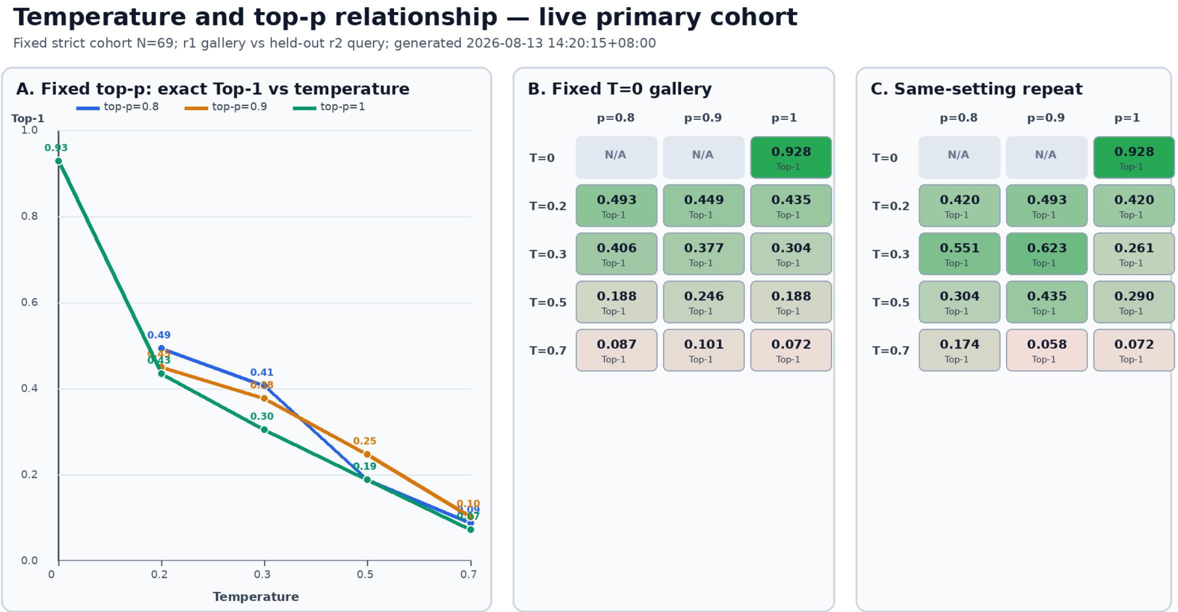

Figure 2: Interim temperature–top-p relationship on the fixed 69-model strict cohort. Panel A fixes top-p and traces exact-model Top-1 as temperature changes. Panel B uses one deterministic $T = 0$ reference gallery for every query cell. Panel C instead uses the matching repeat-1 gallery for each cell. All queries use held-out repeat 2.

Table 2: Exact-model Top-1 against the fixed $T = 0$ gallery ($N = 69$).

Temperature	$p = 0.8$	$p = 0.9$	$p = 1.0$
0	–	–	0.928
0.2	0.493	0.449	0.435
0.3	0.406	0.377	0.304
0.5	0.188	0.246	0.188
0.7	0.087	0.101	0.072

declines from 0.435 at $T = 0.2$ to 0.072 at $T = 0.7$.
- Lower top-p is beneficial at $T = 0.2$ and $T = 0.3$, but the pattern is not monotone at $T = 0.5$ or $T = 0.7$. For example, $p = 0.9$ is best at both of those temperatures.
- The changing top-p ordering is suggestive of a temperature–top-p interaction, but it is not yet a formal interaction test. The final analysis should attach paired model-level uncertainty after recovery.

The same-setting panel is also non-monotone. It is stronger than the fixed gallery at some cells, such as $(T, p) = (0.3, 0.9)$, but remains weak at $T = 0.7$. This indicates that high stochasticity affects both cross-setting robustness and within-setting repeatability.

3 Bounded and jointly bounded decoding conditions

The planned deployment analysis assumes that the exact decoding setting is hidden but an upper temperature bound $T \leq \tau$ may be known. We extend that view descriptively to a joint bound $T \leq \tau$ and $p \leq \pi$.

For every (τ, π) pair, the gallery is the per-model centroid across all eligible repeat-1 cells. The deterministic control is always included because top-p is inactive at $T = 0$. Repeat-2 cells supply the queries. Each decoding cell receives equal weight. Figure 3 shows equal-cell mean Top-1, worst-setting Top-1, and the joint mean grid.

Table 3: Equal-cell mean Top-1 under joint temperature and top-p bounds.

Temperature bound	$p \leq 0.8$	$p \leq 0.9$	$p \leq 1.0$
$\tau = 0.2$	0.768	0.671	0.656
$\tau = 0.3$	0.705	0.661	0.617
$\tau = 0.5$	0.627	0.605	0.564
$\tau = 0.7$	0.475	0.502	0.465

The bounded results reinforce the temperature trend. With $p \leq 0.8$, mean Top-1 declines from 0.768 at $\tau = 0.2$ to 0.475 at $\tau = 0.7$. The worst-setting value declines more sharply, from 0.623 to 0.116. Restricting top-p improves the mean for $\tau \leq 0.5$, but not monotonically at $\tau = 0.7$, where the $p \leq 0.9$ mean is 0.502. Because the gallery centroid and eligible query mixture both change with the bound, these comparisons are deployment summaries rather than a causal decomposition of temperature and top-p.

4 Seed provenance and the four-seed extension

Artifact coverage and seed-clean coverage must remain separate. Table 4 records which existing artifacts can be reused in a four-seed DistDNA study.

Across all 12 stochastic settings, 320 unique successful model–setting artifacts currently have explicit seed 42001 or 42002. Another 176 stochastic artifacts use seed 42. A complete four-seed extension would require 4,800 artifacts, leaving approximately 4,480 to generate after exact-seed reuse.

5 Immediate DistDNA pilot

The first DistDNA result should be a debugging pilot rather than an immediate full-grid commitment. The proposed design fixes one difficult stochastic setting, $T = 0.7, p = 1.0$, and uses

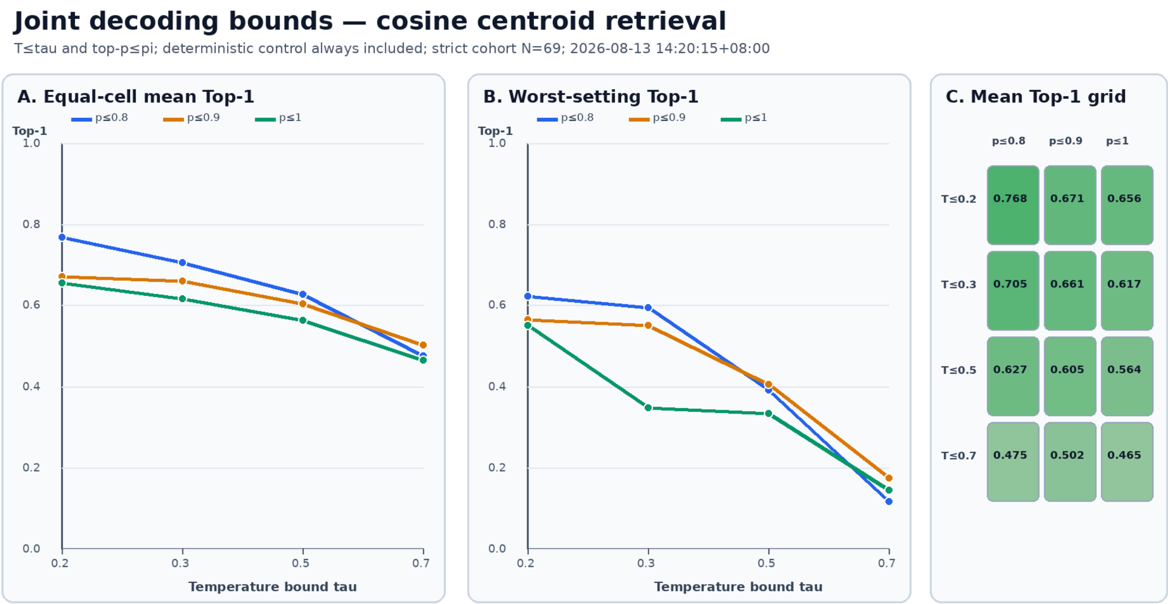

Figure 3: Interim cosine-centroid performance under joint bounds $T \leq \tau$ and $p \leq \pi$ on the fixed 69-model cohort. Panel A reports the equal-cell mean; Panel B reports the worst eligible setting; Panel C shows the mean surface. The deterministic control is included at every bound.

Table 4: Response-generation seed policy.

Source	Recorded seed	Four-seed use
$T \in \{0.2, 0.3, 0.7\}$ backfills	r1: 42001; r2: 42002	Reusable when the exact model, setting, prompts, token budget, and seed match.
$T = 0.5$ control/recovery	r1: 42; r2: 42	Retain for the original grid, but regenerate for seed-clean evidence.
Historical sweep	unknown	Retain for matched comparisons on identical responses, but regenerate for seed-clean evidence.
New extension	42001–42004	Primary four-seed DistDNA artifacts.

four independent seeds for ten models below 0.8B parameters.

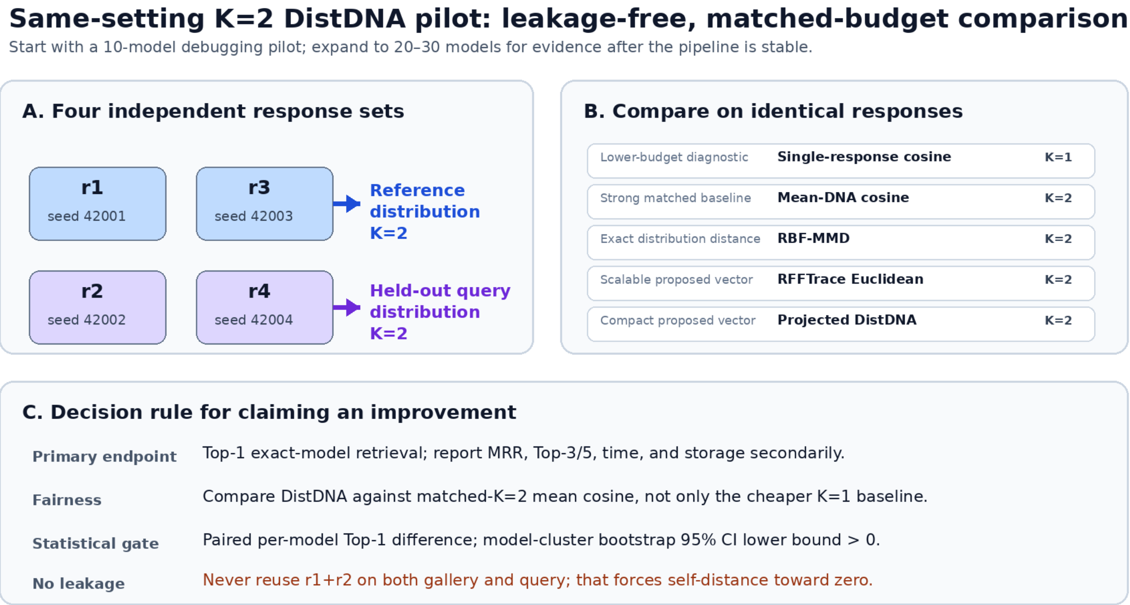

Figure 4: Leakage-free, matched-budget DistDNA pilot. Seeds 42001 and 42003 form the $K = 2$ reference distribution; seeds 42002 and 42004 form the held-out $K = 2$ query distribution. Every $K = 2$ method consumes the same responses.

The ten-model cohort is fixed before DistDNA accuracy is inspected:

Table 5: Proposed small-model pilot cohort.

No.	Model	Approx. scale
1	`ibm-granite/granite-4.0-h-350m-base`	0.35B
2	`EleutherAI/pythia-410m-seed1`	0.41B
3	`EleutherAI/pythia-410m-seed2`	0.41B
4	`EleutherAI/pythia-410m-seed3`	0.41B
5	`unsloth/Qwen2.5-0.5B`	0.50B
6	`unsloth/Qwen2.5-0.5B-Instruct`	0.50B
7	`Ihor/Text2Graph-R1-Qwen2.5-0.5b`	0.50B
8	`robowaifudev/megatron-gpt2-345m`	0.35B
9	`state-spaces/mamba-370m-hf`	0.37B
10	`unsloth/Qwen3-0.6B-Base`	0.60B

The repeated Pythia and Qwen families make the pilot less likely to saturate than an arbitrary set of unrelated models. The parameter labels are advertised model scales and must be checked against model metadata before launch.

5.1 Fair comparison

The pilot reports five retrieval methods:

1. single-response cosine ($K = 1$), as a lower-budget diagnostic;

2. mean-DNA cosine ($K = 2$), as the primary matched-budget baseline;
3. exact RBF-MMD ($K = 2$);
4. RFFTrace ($K = 2$); and
5. compact projected DistDNA ($K = 2$).

The principal contrast is DistDNA minus mean-DNA cosine on the same query models. Beating only the $K = 1$ cosine baseline would not establish a distributional advantage. Top-1 is primary; MRR, Top-3/5, runtime, and storage are secondary. With ten query models, one corrected model changes Top-1 by ten percentage points, so this pilot is suitable for debugging and effect direction rather than a final confidence claim.

6 Execution time and scale-out decision

Historical successful primary artifacts required a median of 9.0 GPU-minutes and a mean of 31.3 GPU-minutes per artifact; recognizable sub-0.8B models had a median of 2.4 minutes but a long retry tail. Table 6 therefore uses operational ranges rather than the median alone.

Table 6: Estimated time after GPUs become available.

Task	Four A100s	Eight A100s
Ten-model, one-setting four-seed generation	1–3 hours	below 2 hours
Checked first DistDNA comparison, including diagnostics	4–8 hours	4–8 hours
100 models, one setting, four seeds	2–4 days	1.5–3 days
100 models, all 12 stochastic settings, four seeds	4–6 weeks	2–3 weeks

At $T = 0.7, p = 1.0$, the four-seed target is 400 artifacts. Approximately 103 explicit 42001/42002 artifacts are already reusable, leaving roughly 297 new artifacts. This one-setting extension should follow immediately after the ten-model pilot passes its leakage and saturation checks. The all-setting extension is a separate multi-week compute commitment and should be gated by the pilot result.

Raw responses remain reusable when RFF dimension, kernel bandwidth, compact projection, or the DistDNA vector is changed. Representation development can therefore continue while the seed-explicit generation queue runs.

7 Implementation and reproducibility

The relevant implementation is:

- distributional features and distances: `src/llm_dna/distributional.py`;
- matched-sample evaluator: `scripts/run_rfftrace_experiment.py`;
- optional SVM evaluator: `scripts/evaluate_rfftrace_svm.py`;
- live recovery and DistDNA design figures: `scripts/plot_primary_status_and_distdna_design.py`; and
- temperature/top-p and bounded-surface analysis: `scripts/analyze_primary_decoding_surface.py`.

The new surface data are saved next to the figures as `primary_decoding_surface.csv`, `primary_bounded_surface.csv`, and `primary_decoding_analysis.json`. Re-running the analysis while recovery is active may change the strict cohort and reported values. The final report must therefore regenerate all counts and accuracy figures after the recovery process exits.